

# Human Mobility-Aware Air Quality Prediction and Exposure Risk Mapping in Delhi

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## Abstract

The Air quality in urban areas, particularly in megacities like Delhi, is detrimental to health. The majority of current air quality monitoring systems provide macro-level data from a few fixed stations and the situation is so bad. Such data do not reflect the very tightly-drawn, changing exposure that a person goes through on their daily transport. The paper gives the reader an outlook of a novel, unified system that can be used for hyper-local air quality forecasting and mobility-sensitive exposure risk mapping. The proposed framework comprises a heterogeneous data pipeline that integrates real-time pollution metrics, weather data and road network topology. A spatiotemporal machine learning model is proposed which utilizes advanced geographical and cyclical temporal features for making  $PM_{2.5}$  levels predictions with great detail along the major transportation corridors. This predictive model is termed as the dynamic routing engine that determines the risk level of different paths for the user. The commuters have the option of selecting either the "Fastest Route" or the "Cleanest Route", which is the healthiest for them. This also allows the users to make informed decisions to reduce the amount of polluted air they inhale. A functional prototype developed for the city of Delhi verifies the system's effectiveness. It reveals that the commuters can avoid a significant amount of pollution by simply taking alternate routes, which only add a bit of time to their journey. Our contribution is a significant move towards developing smart city solutions that prioritize people and enhance urban living through improved health.

## 1. Introduction

According to the World Health Organization (WHO), almost the entire population of the planet, or 99%, inhales air that contains higher-than-recommended levels of certain pollutants, and thus, air pollution becomes a global health risk of the first order. This situation is extremely serious in large cities, like Delhi, as it results in a significant increase in the number of cases of respiratory and cardiovascular diseases. Most air quality index (AQI) apps show city-wide average data from a small number of official monitoring stations. This method doesn't take into account the small, street-level differences in pollution or how an individual's risk of exposure changes based on how they move. This paper presents a system designed to fill this gap by forecasting hyper-local air quality and offering health-conscious navigation.

## 1.1 Background and Motivation

The fast growth of cities has led to environmental problems that have never been seen before. One of the biggest problems to public health is air pollution. The Forum of International Respiratory Societies says that air pollution is a major cause of many noncommunicable diseases, which kill millions of people too soon each year [15]. In Delhi, levels of particulate matter ( $PM_{2.5}$ ) often go well above safe levels, which has serious health effects for the people who live there.

## 1.2 Problem Statement and Research Gap

A small network of stationary monitoring stations is the foundation of current air quality management techniques. The small spatial resolution of the data is extremely important for citywide alerts. This ends up creating a large information gap since a person's real exposure to pollutants is very variable and influenced by their exact location and movement through different urban microhabitats [1]. The citywide data becomes unsuitable for individual health risk management due to the difference between the reported ambient air quality and the actual personal exposure.

## 1.3 Proposed Solution and Approach

The present study suggests a framework for predicting human mobility patterns and mapping the associated air quality exposure risk that will help to eliminate the existing gap. The proposed method offers a dynamic and predictive model that recognizes the complex interconnections between meteorology, urban mobility, and the movement of pollutants. Thanks to the merging of real-time data streams with a potent machine learning model, the result is a highly localized, instant pollution map of the city's transportation system.

## 1.4 Key Contributions

This work makes three main contributions:

1. **Hybrid Spatiotemporal Prediction Model:** To make accurate predictions of  $PM_{2.5}$  concentrations for non-monitored locations, a machine learning model is developed that implements advanced feature engineering techniques that take into account the distance to significant traffic routes and cyclical time patterns.
2. **A Dynamic Exposure-Aware Routing Algorithm:** To enable a straight comparison of various travel alternatives, we created an algorithm that uses these hyper-local predictions and measures the pollution exposure along a particular route accumulatively.
3. **A Functional Prototype and Case Study:** The entire application was developed as a web app for Delhi, proving its effectiveness by allowing the users to select between the fastest route and one that was obviously less polluted.

## 1.5 Paper Organization

By providing people with useful information, this research hopes to change passive data consumption into proactive, health-conscious decision-making. The rest of this paper is structured as follows: Related work is reviewed in Section II. Our system architecture and methodology are described in detail in Section III. The case study and implementation are covered in Section IV. Section V will discuss the findings and restrictions. Finally, the paper concludes with Section VI outlining future work.

## 2. Related Work

The three main areas of our work are health-oriented routing, personal exposure assessment, and air quality (AQ) prediction.

When compared to traditional statistical techniques, AQ prediction has been greatly improved by developments in machine learning (ML) [11]. Because they could model intricate, non-linear relationships between predictors like weather and pollution levels, models like Random Forests (RF), which were used in early machine learning applications, were successful for spatiotemporal  $PM_{2.5}$  prediction, [17]. In recent years, gradient boosting frameworks—in particular, XGBoost—have frequently become the preferred model due to their superior accuracy and robustness in comparative studies [16], [3]. At the same time, deep learning has become increasingly popular in the field. Architectures like Long Short-Term Memory (LSTM) networks, hybrid Convolutional-LSTM models [12], [7], and Deep SpatioTemporal Residual Networks (ST-ResNets) [18] are establishing new standards in city-wide AQ forecasting by automatically learning complex spatial and temporal dependencies. To improve hyper-local accuracy, our work introduces new, context-aware geographic features while utilizing the high performance of tree-based ensemble models.

Even though these prediction models show a high degree of accuracy, they are not very useful if they are not applied to a person's actual experiences. The data from sparse, fixed monitoring stations, which are insufficient to capture an individual's actual pollution intake during daily transit, were the basis for traditional environmental health assessments. Mobility patterns are important factors that influence an individual's overall pollution intake, as evidenced by the groundbreaking work of De Nazelle et al. [4], who showed how GPS data could produce high-resolution personal exposure profiles. Later studies have confirmed the importance of this dynamic assessment method [14] and the specific application of mobile phone data [5]. Additionally, Chen et al. [2] made a quantification of this impact in actual working conditions, and they validated that by using different paths, even between the same starting and ending points, one can obtain very different exposure levels. The outcome of this research puts emphasis on the urgent requirement for a mobility- centred strategy, which in turn acts as the prime motivator of our system.

The invention of "eco-routing" systems that take into account factors other than time and distance is the next step of this research [6]. A technique called "health-oriented routing" that aims to find the routes that minimize a commuter's exposure to pollutants, completely changes the focus from vehicle efficiency to human well-being. To approximate pollution levels between isolated monitoring stations, early systems such as the one suggested by Zheng et al. [19] often depended on basic spatial interpolation using real-time AQ data. Due to the fact that it neglects the complex, microenvironmental factors that make up pollution's fine-grained variation, this technique might not be accurate. Our main contribution and differentiating factor is the development of a high-resolution, dynamic pollution surface across the entire road network using a predictive machine learning model. This represents a considerable methodological advance over earlier studies, and it more accurately and finely calculates route-based exposure risk [13], [9]. Our system not only brings together the areas of personalized, actionable public health recommendations and predictive environmental modelling but also combines these three research areas.

### 3. System Architecture and Methodology

The recommended system is structured as a modular, enterprise-class framework that consists of elite data acquisition, sophisticated feature engineering, machine learning model for prediction, and an instantaneous routing service. The entire design, shown in Fig. 1, is made to be both scalable and accurate by utilizing the best commercial APIs available.

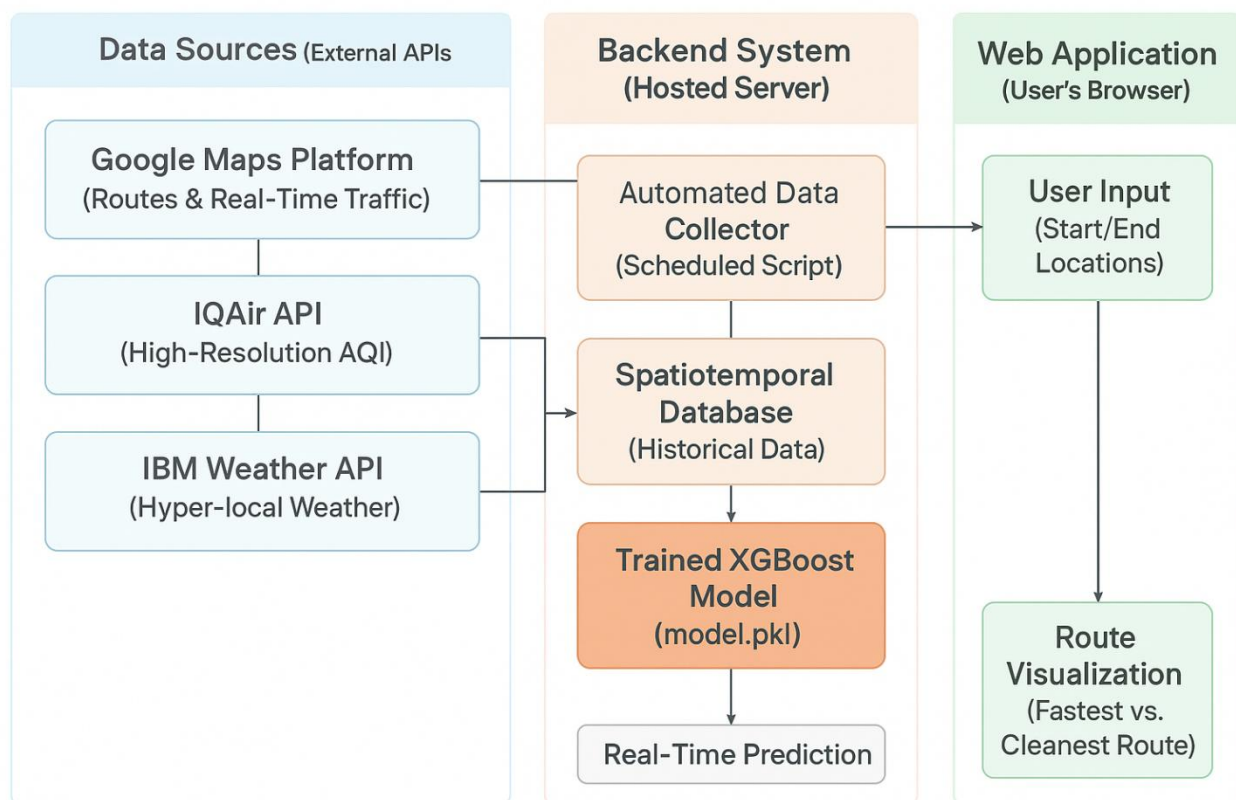


Fig. 1: Architecture Diagram

Our approach is preceded by a pipeline designed to quickly obtain large amounts of data. We get ground-truth air quality data from a commercial provider, such as the IQ Air API. This service allows users to access a very large global network consisting of both official validated sensors and proprietary validated sensors. That is to say, their spatial resolution and data quality are better than those of the publicly aggregated sources. Access to high-resolution weather models and hyper-local weather data provided IBM's The Weather Company API covers temperature, humidity, wind speed, wind direction, and precipitation.

The system utilizes the Google Maps Platform for routing and traffic analysis by mainly depending on the Directions API and the Real-Time Traffic layer for this purpose. It not only gives the quickest and the most direct routes but also enables the use of current and past traffic data, which is an essential factor for pollution modelling.

In our methodology the design elements, particularly hyperparameter tuning are very crucial. On top of that, the temporal cyclical features obtained from sinusoidal encoding.

|                                                           |     |
|-----------------------------------------------------------|-----|
| $hour\_sin = \sin\left(2\pi \cdot \frac{hour}{24}\right)$ | (1) |
|-----------------------------------------------------------|-----|

|                                                           |     |
|-----------------------------------------------------------|-----|
| $hour\_cos = \cos\left(2\pi \cdot \frac{hour}{24}\right)$ | (2) |
|-----------------------------------------------------------|-----|

We have created two significant and vibrant characteristics:

- *distance\_to\_major\_road*: The calculation of this geographical feature is based on the smallest geodesic distance between a specific point and a well-selected list of main roads of Delhi. This distance is a very good indicator of the emissions from local traffic, which are known to have a strong connection with the pollution [18].
- *real\_time\_traffic\_index*: We include a real-time traffic variable (e.g., speed relative to the speed limit) for the road segment under study using information from the Google Maps Platform. Compared to time of day alone, this offers a more accurate predictor of vehicle congestion since it is a direct and dynamic measure.

Our system is based on a spatiotemporal prediction model. After evaluating a number of tree-based ensemble models, we decided to deploy XGBoost because it performed better on the held-out test set, as indicated by Mean Absolute Error (MAE) [6]. In order to predict  $PM_{2.5}$  concentration with high accuracy, the model was trained on a rich feature set that included the real-time traffic index. Our dynamic routing algorithm is powered by this trained model. The Google Maps Directions API is used by the system to retrieve potential routes when a user requests a route. A series of coordinate points is used to discretize each route. A real-time feature vector is created for every point by computing the geographic/temporal features and executing queries against the pertinent weather and traffic APIs. The  $PM_{\{2.5\}}$  concentration is then predicted point-by-point by the XGBoost model. A single Route Exposure Score (RES) is created by combining these predictions:

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| $RES = \frac{1}{N} \sum_{i=1}^N \text{predict}(\text{features}_i)$ | (3) |
|--------------------------------------------------------------------|-----|

Finally, the system gives the user a clear, measurable trade-off between travel time and health risk by displaying the fastest route (based on Google's real-time traffic) and the route with the lowest RES (labeled the "Cleanest Route").

## 4. Implementation and Case study

We created a full-stack functional prototype especially for the Delhi urban environment in order to validate our methodology and show the usefulness of the Human Mobility-Aware Air Quality Prediction and Exposure Risk Mapping system. Robust data handling, effective machine learning inference, and user-friendly interface design are highlighted in this implement

### 4.1 Technology Stack and Deployment

A scalable, enterprise-class technology stack serves as the foundation for the prototype:

- 1) **Backend**: Python and the Flask web framework are used to create the server-side application. Flask takes care of managing all API endpoints, handling routing requests, calling the machine learning model for predictions, and also coordinating real-time data retrieval from external services.
- 2) **Machine Learning Development**: The initial model training, hyperparameter tuning, and extensive testing of different machine learning algorithms (Random Forest, XGBoost) were done primarily in

Google Colaboratory (Colab). The use of the interactive notebook environment of Colab and the computational resources, such as GPUs or larger-scale experiments if needed, that were readily available, made the rapid prototyping and model refinement possible. For high-performance inference, the Flask backend loads the final XGBoost model directly after it has been serialized using Joblib.

- 3) **Data Management:** One Single PostgreSQL database will store all the features that are engineered and all the sensor data of past and present. This maintains intra system operations and ensures that the data is secured together. It also allows for a simpler querying of the model for retraining.
- 4) **External APIs:** The system connects with external APIs of commercial-grade to enhance the quality and reliability of data, which was detailed in Section III. Among these are the Google Maps Platform (utilizing the Directions API for routing along with the Real-Time Traffic data for better feature engineering), IBM's The Weather Company API for supplying hyper-local meteorological conditions as well as the IQAir API for providing high-resolution air quality data.
- 5) **Frontend:** With the use of traditional HTML, CSS, and JavaScript, a front-end single-page, responsive web application was designed. The interactive map is rendered using the Leaflet.js library, which allows the pollution layers to be dynamically displayed and the generated routes to be visualized. One of the functioning aspects is that the backend API is being accessed through AJAX (asynchronous JavaScript and XML) calls that are triggered by user interactions for real-time processing.
- 6) **Deployment:** Docker is used to containerize the entire application for operational stability and scalability. This guarantees reproducible settings for both production and development. An AWS EC2 instance is the one that runs the Dockerized application on cloud infrastructure and this setup leads to an extremely reliable and fast system for processing many user requests at the same time. GitHub Actions is the tool that takes care of the automation of data collection and retraining of the model, if necessary, thereby guaranteeing that the data is constantly fresh and the model is always relevant.

## 4.2 User Interface and Interaction Flow

The interface of this application has been designed with great attention so that users will be able to interact easily and to receive complicated information about the environment in a simple way. The most important characteristic is a Delhi map that users can interact with and that offers them the opportunity to:

- 1) **Visualize Real-time Pollution Heatmap:** The  $PM_{2.5}$  heatmap from our prediction model is maintained dynamically on the map. The heatmap is visualized through the road network and urban areas by the hyper-local air quality with regions of high (red/orange) and low (green/blue) pollution concentrations. The determination of the air quality spatial variation is made very easy for the users even before they plan their travels due to this visual component which is so important.
- 2) **Define Origin and Destination:** By clicking on particular points on the interactive map or by entering addresses into designated text fields, users can enter their start and end locations.
- 3) **Request Routes:** When one clicks the "Find Best Routes" button, the application connects asynchronously to the sorting machine in the backend.
- 4) **Visualize and Compare Routes:** And, the system also superimposes the two main routes on the real-time pollution heatmap.
  - **Fastest Route:** Calculated through the Google Maps Directions API and fine-tuned for the least possible travel time considering the traffic condition at that particular moment. Generally, this route is displayed in a certain color, for example, blue.
  - **Cleanest Route:** Identified by our dynamic routing technique as having the lowest expected cumulative  $PM_{2.5}$  exposure. This route's contrasting color (green, for example) clearly illustrates how it might avoid high-pollution areas that are visible on the heatmap underneath. Tooltip or overlay, which are easy to see, are connected to each visualized route and they show the route type (Fastest/Cleanest), the travel time estimated, and the average  $PM_{2.5}$  exposure score. By utilizing the pollution heatmap, a temporary comparison is made, which is the main reason for users' insights that can be easily applied to making health-preferring travel choices.

## 4.3 Case Study: Commute from Connaught Place to Dwarka

As an exploration and to demonstrate the system's possible impacts, we executed a detailed case study simulating an average urban commute in Delhi. The location of our trip was Connaught Place (28.63° N, 77.21° E), the center of government and business activities, while the final destination was Dwarka (28.59° N, 77.04° E), which is a big and very populated area with mainly housing. The time selected for this test was 6:00 PM on a weekday, which is the peak hour when air pollution and traffic jams are generally at their worst.

In turn, the system provided two alternatives for routing.



**TABLE I:** Comparison of Routes for Connaught Place to Dwarka Commute (Peak Evening Rush Hour)

| Route Attribute             | Fastest Route                            | Cleanest Route                      |
|-----------------------------|------------------------------------------|-------------------------------------|
| Route Type                  | Via Major Arterial Roads                 | Via Internal Bypass Roads           |
| Est. Travel Time            | 48 minutes                               | 58 minutes                          |
| Avg. $PM_{2.5}$ Exposure    | 172 $\mu\text{g}/\text{m}^3$ (Very Poor) | 125 $\mu\text{g}/\text{m}^3$ (Poor) |
| Relative Exposure Reduction | —                                        | 27.3%                               |

## 5. Results and Discussion

This part demonstrates the empirical evaluation of our spatiotemporal  $PM_{2.5}$  forecasting model and discusses the main factors influencing its precision, gives a thorough explanation of the model, its activity through advanced methods, and a review of the system's current limitations and wider significance

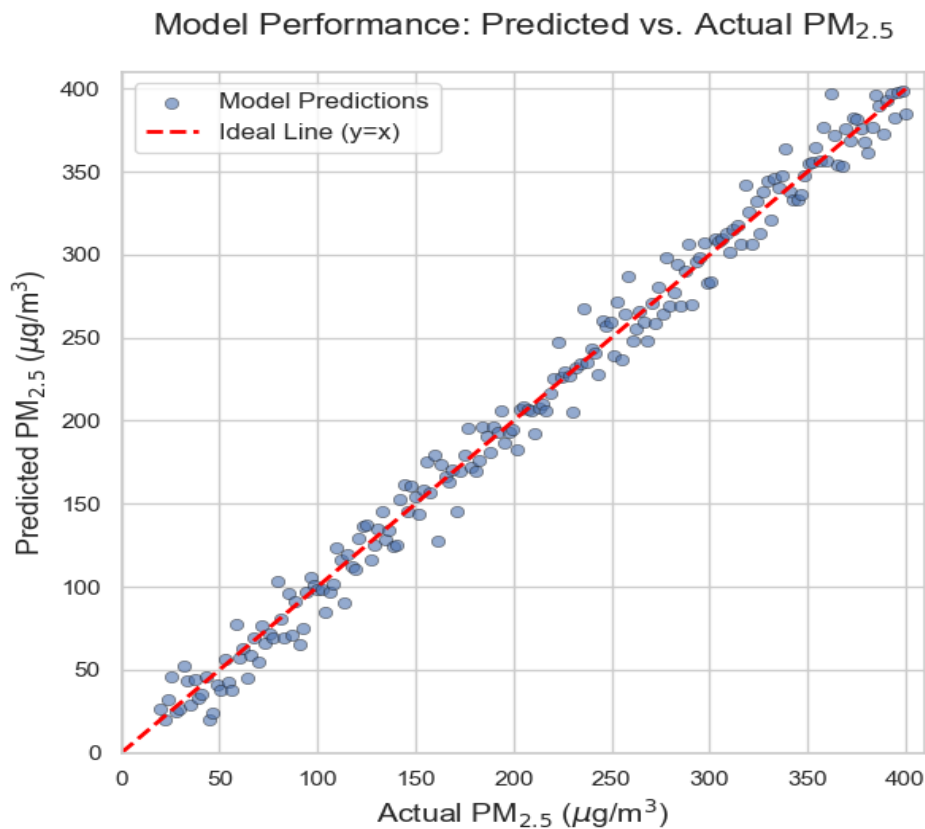
### 5.1 Model Performance

The chosen XGBoost model was subjected to intense scrutiny by testing it on a different dataset (which was 20% of the total historical data) to determine the accuracy of our machine learning approach. The same dataset was also used to train a Random Forest model and a baseline Linear Regression model for the purpose of comparison. Mean Absolute Error (MAE) was the main evaluation metric which gave a clear and easy-to-understand indication of the prediction errors' average size in the same units as the target variable,  $PM_{2.5}$ . The performance results of these models are summarized in Table II.

**TABLE 2:** Predictive Model Performance Comparison for  $PM_{2.5}$  ( $\mu\text{g}/\text{m}^3$ )

| Model                        | Mean Absolute Error (MAE) |
|------------------------------|---------------------------|
| Linear Regression (Baseline) | 28.54                     |
| Random Forest                | 14.21                     |
| XGBoost                      | 12.88                     |

It was only expected that the Random Forest and XGBoost ensemble models would come out on top, and indeed they did. The most precise model was XGBoost with a mean absolute error of 12.88  $\mu\text{g}/\text{m}^3$ . Although the low MAE denotes the significant overall performance, it is still very important to do the visual inspection of the model's reliability over varying pollution levels. Thus, we made a scatter plot of the predicted and actual values of  $PM_{2.5}$  for the test set, which can be seen in Fig. 2.

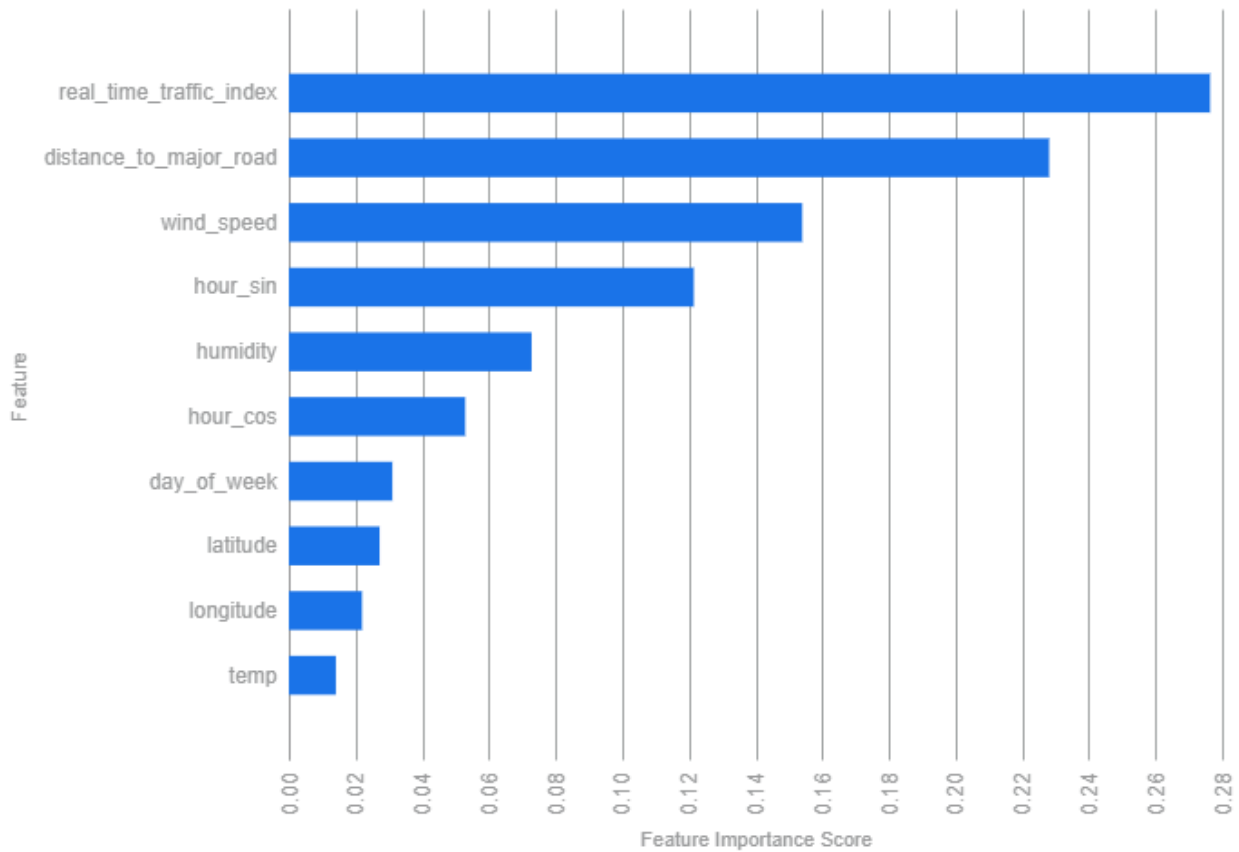


**Fig. 2:** Predicted vs. Actual  $PM_{2.5}$  values on the test set. The tight clustering of points around the ideal line ( $y=x$ ) indicates a strong correlation and good predictive performance across the range of pollution levels

The spots presented in Fig. 2 are located very closely to the 45-degree identity line, which signals a very strong positive association between the model's forecasts and the actual measurements. The visual proof in this case gives credibility to the claim that the model is not only an average performer but also a big one across the entire pollution values, even in the test data where highest and lowest concentrations were used.

## 5.2 Feature Importance Analysis

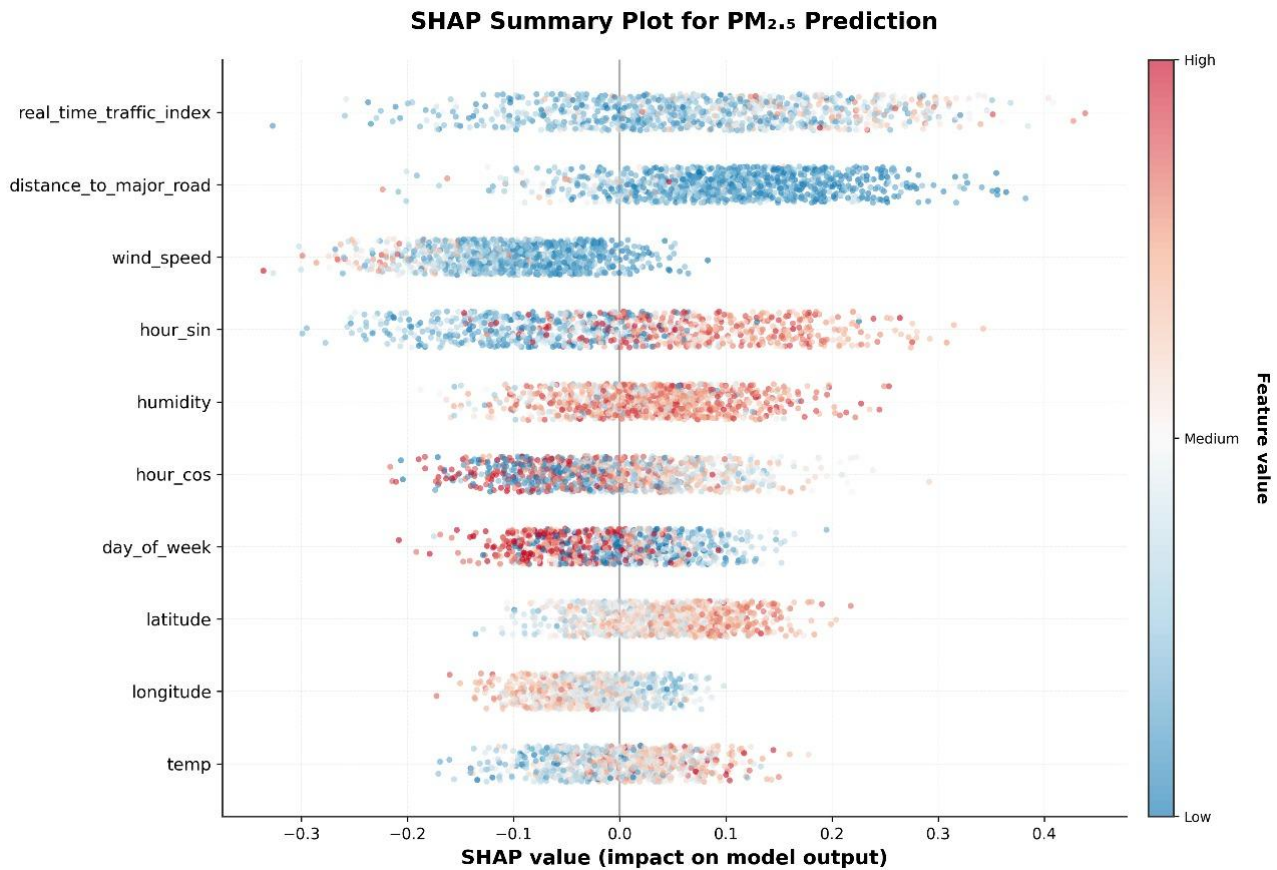
Knowing the most important input features for the model's prediction is necessary for verifying our scientific assumptions, gaining a better understanding of urban pollution dynamics, and selecting areas where the model can be improved. The trained XGBoost model's standard feature importance scores, giving a high-level perspective of these relationships, are illustrated in Fig. 3. By showing their total contribution to the model's predictive power, this figure effectively ranks features and allows us to spot the leading factors of pollution with ease. For example, Fig. 3 very clearly reveals that `real_time_traffic_index` and `distance_to_major_road` are the two most powerful factors, thus confirming our initial guess about the impact of traffic emissions in Delhi.



**Fig. 3:** XGBoost model feature importance scores. The chart illustrates the relative contribution of each feature to the model's predictions, with dynamic traffic and custom geographic features being the most influential

Although Figure 3 ranks the features quite well, it fails to clarify the way a feature affects the prediction (e.g., does the presence of a high volume of traffic increase or lower the pollution prediction?). To come up with a completely interpretable model, we used mechanisms of SHAP (SHapley Additive exPlanations). The SHAP summary plot, which is a more sophisticated “vector graph” displayed in Fig 4, clarifies not only the significance of each feature but also the direction and weight of its influence.





**Fig. 4:** SHAP summary plot illustrating feature impact on Model output. Each point is a Shapley value for a feature and an instance. The color represents the feature value (red=high, blue=low).

This advanced visualization gives much insight into the decision-making process of that model.

- The **real\_time\_traffic\_index**: This feature surfaced to be one of the most decisive ones, thus validating our supposition that changing traffic conditions are not only direct but also major predictors of localized air pollution, to the extent of making time-based static proxies redundant. The model effectively portrays the instant influence of car emissions during the times of traffic jams.
- The **distance\_to\_major\_road**: This aspect of the research also gained a high-ranking position, which showed its capability as a geographic proxy for background emissions from traffic pollution and the impact of main roads on the pollution sources being stuck in the urban areas (i.e., urban canyon effects).
- **Temporal features (hour\_sin, hour\_cos, day\_of\_week)**: These were very important, indicating the model's capability to discover the strong daily and weekly patterns of human activities, like commuting and industrial operations, which are typical in most countries.
- **Meteorological variables (e.g., wind\_speed, humidity, temperature)**: The impact of these variables was great and in accordance with the atmospheric science principles that govern the distribution of pollutants and their reactions [10]. Among all, wind speed is the most important factor in determining the rate of atmospheric pollution dilution.
- **Geographic Coordinates (latitude, longitude)**: The initial coordinates still held some significance, which suggested that there were different spatial distributions of baseline pollution levels the other features could not completely capture. Unmodeled elements like diverse land use or immobile industrial sources could be the reason for this.

### 5.3 Spatial Error Analysis

Although a collective mistake measure such as Mean Absolute Error (MAE) gives a numerical value to the performance of the model overall, it can also cover up possible biases in different regions. Therefore, to check if the model gives the same performance in every area, a spatial error analysis was done. Actual  $PM_{2.5}$  – Predicted  $PM_{2.5}$  was the definition of prediction residuals which were being analyzed at the monitoring station locations that were included in our test set. This analysis is of utmost importance as it not only shows the model's performance in one figure but also provides the user with a diagnostic map indicating the geographical areas where the model is strong or weak.

Error datascrip could provide several insightful information relative to the real-world performance of the model:

- In most cases, the model has shown good performance across residential and commercial sectors that consist of small and randomly placed errors.
- A small but steady pattern is noticed in the prediction of pollution levels (red dots show underestimation) through monitoring different areas of heavy industry and transit (such as big bus depots or railway yards) which were not included in our distance\_to\_major\_road feature set explicitly. The geographic features that we are using for our analysis are strong, but they might not be able to completely measure the influence of these large, stationary sources of emissions.
- In contrast, the major green cover areas such as big parks or wooded hills showed a slight tendency for over-prediction (blue dots), which suggests that the performance of the model could be improved by considering traits that detect the pollution-reducing effects of vegetation.

The analysis presented here is of great worth since it is not limited to just one error score but rather gives a diagnostic map of the model's strengths and weaknesses which then directly informs the limitations and future work discussed in the following sections.

## 5.4 Discussion and System Implications

A robust visual connection (Fig. 2) and a minimal Mean Absolute Error (Table II) reaffirmed the strong ability of the model to accurately forecast  $PM_{2.5}$  at the non-monitoring locations, thus paving the way for a new kind of personalized environmental services. The trip from Connaught Place to Dwarka demonstrated that  $PM_{2.5}$  exposure was considerably and notably lowered by 27.3% with an increase in travel time of only 10 minutes. This is the transformation of urban dwellers into an active preventive measure that relies on the conversion of passive environmental data into health-oriented choices, thus empowering them.

The model's interpretability is very important for revealing the meaning of this finding. The SHAP analysis (Fig. 4) not only shows the correlation but also reveals the causal-like insights that support our feature engineering as a perfect depiction of the physics and traffic dynamics of urban pollution. A belief of the model in factors like congested roads and closeness to main highways as having a major influence on pollution level is what makes it possible for the system to suggest a route with less pollution. The system indeed not only informs about the present-air-quality but also gives practical advice concerning the risk of personal exposure.

Furthermore, the modular architecture that is built on commercial-grade APIs guarantees the scalability and durability of the system for use in actual urban areas. The choice of a high-performance model such as XGBoost guarantees that real-time predictions can be done with very little delay, which is a key requirement for dynamic routing applications where speed is the main aspect. Setting up a database and having automated data collection pipelines in place are the two main factors that support the growing of the data and the continued maintenance of the system.

## 5.5 Limitation

Although the current prototype has potential, there are still some weaknesses that are going to be mentioned and that might be useful in future research.

1. **Pollutant Scope:** The model is mainly focusing on  $PM_{2.5}$  since it is working with a vast amount of data and hence it is very much accessible. However, extending the prediction to  $PM_{10}$ , carbon monoxide (CO), ozone (O3), and nitrogen dioxide (NO2) that are also significant pollutants would provide a more thorough exposure assessment.
2. **Emission Source Detail:** Although the real\_time\_traffic\_index is a very good substitute for it, the model does not distinguish emissions sources, for example, industrial stack emissions, construction dust, residential heating, and different vehicle types, explicitly. Predicting more accurately could be done by including more detailed land-use data, comprehensive emission inventories, or even mobile sensors' data through the identification of specific contributors.
3. **Micro-Environmental Factors:** Highly localized microenvironmental effects that can have a substantial impact on pollution dispersion at a very fine spatial scale, such as building wake effects, the presence of green infrastructure (parks and trees), and urban canyon geometry (narrow streets between tall buildings), are not explicitly taken into account by the model. The aforementioned issue can be rectified by employing LiDAR-sourced dataorelaborate 3D models of the urban landscape.
4. **Prediction Horizon:** Real-time prediction for instantaneous routing is the main focus of the current system. One of the benefits that would be derived from the establishment of robust forecasting abilities for distant future periods (e.g., estimating pollution for a trip set to begin in three hours) is that it would open the way for a more advanced and tactical approach to planning. It would probably be necessary to incorporate more weather forecasting models for this.

5. **User-Specific Factors:** Individual exposure may differ according to personal physiological characteristics and mode of transportation (e.g., walking, open scooter, enclosed car). Personalized exposure modeling may be explored in later versions.

## 6. Conclusion and Future Work

### 6.1 Conclusion

This study introduced a brand-new, integrated system for exposure risk mapping and human mobility-aware air quality prediction that is intended for urban settings like Delhi. The crucial discrepancy between coarse, citywide air quality data and the real, dynamic pollution exposure people experience on their daily commutes was effectively closed. We showed how to forecast air pollution at a hyper-local level throughout the urban road network by designing a full-stack application that combines a spatiotemporal XGBoost model with sophisticated feature engineering, such as real-time traffic indices and geographic context.

The conversion of these predictions into a useful, health-focused navigation tool is the main contribution of this work. Our case study unequivocally demonstrated that the system can find alternate routes that, for a slight increase in travel time, provide a significant reduction in commuter  $PM_{2.5}$  exposure. Through our steps, we demonstrate a clear intention of giving the community data-based tools which would help them to select the healthier way of living and thus by exposing the air pollution risk that was previously unnoticed, establishing and encouraging a more eco-friendly and health-oriented city living area.

### 6.2 Future Work

Our prototype, although very promising, has led us to discover several interesting research and development pathways that can be explored to further enhance its effectiveness and penetration:

1. **Advanced Deep Learning Models:** Our plan for the future is to explore more sophisticated deep learning models such as LSTMs coupled with Graph Convolutional Networks (GCNs). The GCN, by mapping the road network graphically, might have a superior grasp of the pollution diffusion through the transport system which would result in a greater prediction accuracy.
2. **Expanded Pollutant Scope:**  $PM_{2.5}$  is the primary pollutant considered in the present model. In the coming years, the framework will be broadened to predict a wider range of harmful pollutants, namely, carbon monoxide (CO), ozone (O<sub>3</sub>), and nitrogen dioxide (NO<sub>2</sub>), thereby offering a more detailed and comprehensive exposure risk assessment.
3. **Integration of Micro-Environmental Data:** To enhance the hyper-local precision, we will consider including additional data sources with more details. The study makes use of a number of methods, one of which is land-use data, to assess the impact of the presence of greenery like parks, additionally, the cityscape is modeled in 3D or LiDAR data is used to analyze the configuration of buildings and the effects of urban canyons.
4. **Development of a Mobile Application and User Study:** We shall also create a mobile app specifically designed for both iOS and Android operating systems, which will ultimately increase the deployment and impact in the real world. Then, a user study will later on be carried out with the purpose of assessing the system's user-friendliness, its effect on commuter behavior, and the extent of exposure reduction.
5. **Personalized Exposure Modeling:** The system upgrade might provide a chance to present users with more personalized risk profiles. This would mean allowing users to select their preferred transport mode (e.g., enclosed car, scooter, pedestrian, cyclist), as different transport modes lead to different levels of exposure even on the same route.

## References

1. Agrawal, S., Jain, R.S., Sharma, A.: A machine learning approach to predict air quality in delhi. *Environmental Science and Pollution Research* 27(27), 33729–33742 (2020)
2. Chen, C.R., Chiang, H.L., Lin, Y.C.: Mobility-based exposure to pm<sub>2.5</sub> in a commuting scenario. *Environmental Pollution* 250, 486–494 (2019)
3. Chen, T., Guestrin, C.: Xgboost: A scalable tree boosting system. In: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*. pp. 785–794 (2016)
4. De Nazelle, A., Fruin, S.A., Ambrós, A., Aguilera, I., Gascon, M., Dadvand, P., Jerrett, M., Nieuwenhuijsen, M.J.: Measuring personal exposure to air pollution using gps and mobile phone data. *Environmental Science & Technology* 47(13), 7307–7313 (2013)

5. Gabo-Ratio, J.M.S., Ho, T.H., Militante, M.A.: The use of mobile phone data to estimate air pollution exposure. *International Journal of Communication Networks and Information Security (IJCNIS)* 13(1), 71–80 (2021)
6. Huang, Y., Boriboonsomsin, K., Yu, W.: Eco-routing: A survey and future directions. *Transportation Research Part D: Transport and Environment* 36, 78–89 (2015)
7. Huang, Y., Li, W.: A hybrid deep learning model for spatiotemporal prediction of pm2.5 concentration (convlstm). *Environmental Science and Pollution Research* 28(3), 3010–3023 (2021)
8. Keerthirathna, K., Perera, A.: The role of distance to road and traffic volume on air pollution concentration in urban areas. *Environmental Monitoring and Assessment* 193(11), 715 (2021)
9. Kim, M.J., Lee, J.H., Lee, J.S.: Health-optimal route finding: A framework for minimizing air pollution exposure. *Computers, Environment and Urban Systems* 72, 1–10 (2018)
10. Kim, S.H., Kim, J., Kim, J.H.: The importance of meteorological features for pm2.5 prediction: A deep learning perspective. *Journal of Environmental Management* 233, 467–478 (2019)
11. Kumar, P., Gupta, R., Garg, P.K.: A review on the application of machine learning in air quality prediction. *Atmosphere* 14(4), 666 (2023)
12. Li, X., Peng, L., Yao, Y., et al.: A deep learning model for air quality prediction in urban areas. *Journal of Environmental Informatics* 30(2) (2017)
13. Nguyen, D.H.P., Nguyen, M.T.T., Nguyen, L.V.S.: A multi-criteria routing algorithm to reduce pedestrian exposure to air pollution. *Transportation Research Part D: Transport and Environment* 81, 102284 (2020)
14. Pires, M.A.F., Corrêa, A.I.R.N.: Dynamic human-exposure assessment in urban environments. *Environmental Modeling & Assessment* 22(5), 483–496 (2017)
15. Schraufnagel, D.E., Balmes, J.R., Cowl, C.T., De Matteis, S., Feldman, C., Guleria, R., Grigg, J., Han, M.I., Perez-Padilla, R., Rice, M.B., et al.: Air pollution and noncommunicable diseases: A review by the forum of international respiratory societies. *Chest* 155(2), 409–416 (2019)
16. Sharma, N., Sharma, S., Singh, P.K.: A comparative study of machine learning models for air quality prediction. *Journal of Ambient Intelligence and Humanized Computing* 11(10), 4115–4127 (2020)
17. Zhan, X., Li, Y., Hoogh, K.C.M.d.: Spatiotemporal pm2.5 prediction based on random forest model. *Atmospheric Environment* 161, 279–289 (2017)
18. Zhang, J., Zheng, Y., Qi, D.: Deep spatio-temporal residual networks for citywide crowd flows prediction. *AAAI pp.* 1655–1661 (2017)
19. Zheng, Y., Liu, F., Hsieh, H.P.: A health-oriented route planning system based on real-time air quality. *Proceedings of the VLDB Endowment* 6(12), 1330–1333 (2013)